Lecture 7

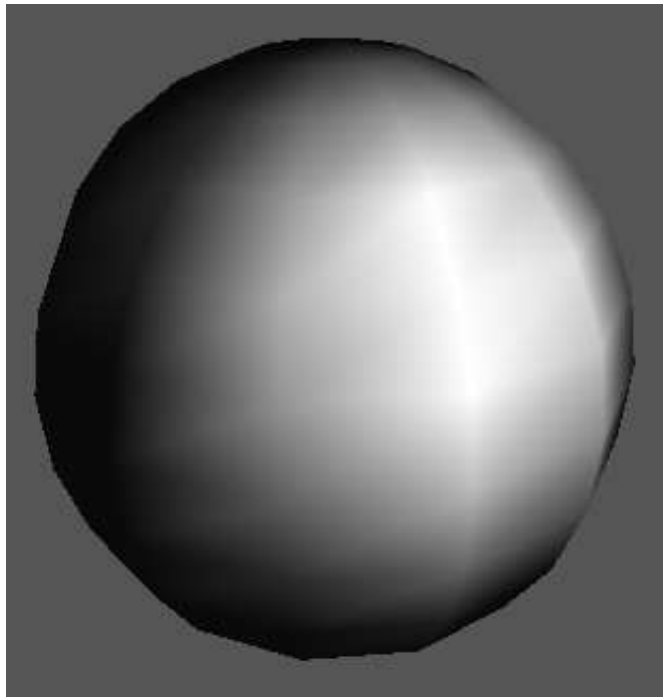
Lighting and Shading

Lighting vs. Shading

- Commonly misused terms.
- What’s the difference?
- Lighting designates the interaction between materials and light sources (in Physics).
- Shading is the process of determining the color of a pixel (i.e. Computer Graphics).
  - Usually determined by lighting.
  - Could use other methods: random color, NPR, etc.

“Lighting”

- Two components:
  - **Lighting Model** or **Shading Model** - how we calculate the intensity at a point on the surface
  - **Surface Rendering Method** - How we calculate the intensity at each pixel

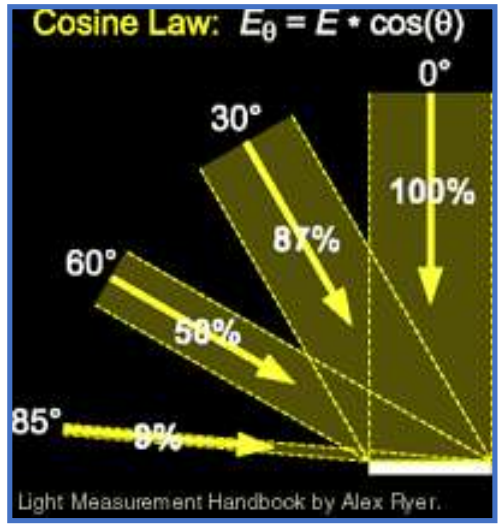


Lighting Models

- Lambert.
  - Purely diffuse surfaces.
- Phong.
  - Adds perceptually-based specular term.
- Torrance-sparrow:
  - Provides a physical approximation.

Lambert's Cosine Law

- The reflected luminous intensity in any direction from a perfectly diffusing surface varies as the cosine of the angle between the direction of incident light and the normal vector of the surface.
- Intuitively: cross-sectional area of the "beam" intersecting an element of surface area is smaller for greater angles with the normal.



Phong Lighting Model

- Phong adds specular highlights.
- His original formula for the specular term:
  - $W(i)[\cos s\ ]^n$
    - s is the angle between the view and specular reflection directions.
    - *"$W(i)$ is a function which gives the ratio of the specular reflected light and the incident light as a function of the the incident angle i."*
      - Ranges from 10 to 80 percent.
    - *"n is a power which models the specular reflected light for each material."*
      - Ranges from 1 to 10.

Torrance-Sparrow Model

- Introduced by Torrance and Sparrow in 1967 as a theoretical model.
- Introduced to CG community by Blinn in 1977.
  - same paper as “Halfway Vector” (Blinn-Phong).
- Attempts to provide a more physical model for specular reflections from real surfaces.
  - Points out that intensity of specular highlights is dependent on the incident direction relative to normal.
  - Phong attempted to model this with $w(i)$ factor.

Shading

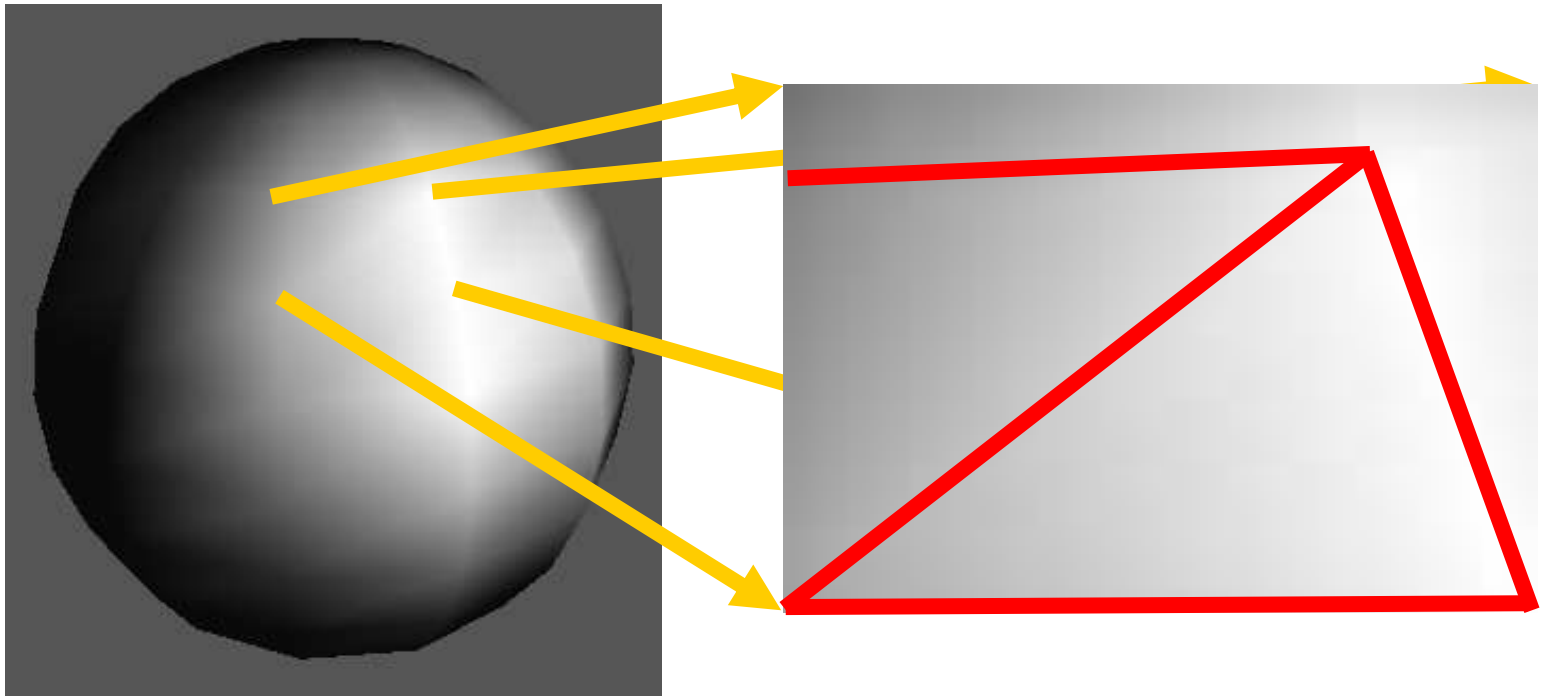
- Shading is how we “color” a triangle.
- Three shading techniques:
  - Constant Shading
    - Calculate one lighting calculation (pick a vertex) per triangle
    - Color the *entire* triangle the same color
  - Gouraud Shading
    - Calculate three lighting calculations (the vertices) per triangle
    - Linearly interpolate the colors as you scan convert
  - Phong Shading
    - While you scan convert, linearly interpolate the normals.
    - With the interpolated normal at each pixel, calculate the lighting at each pixel

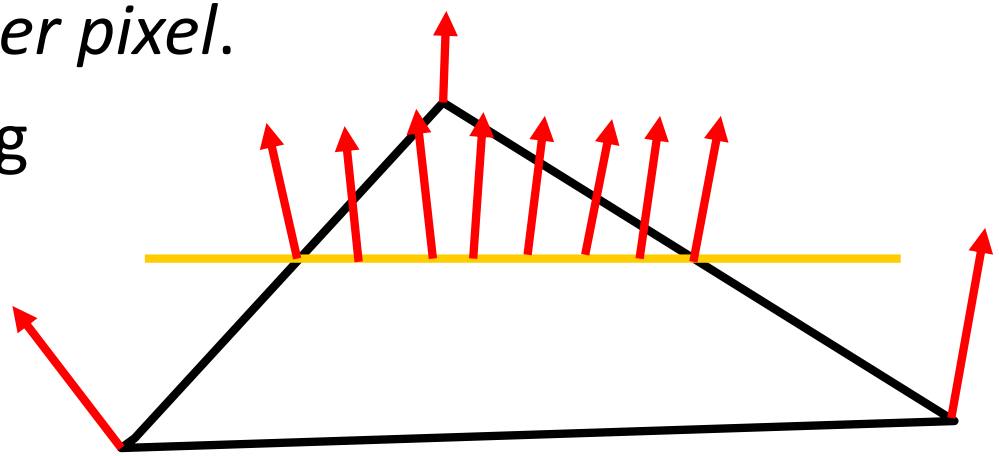
Constant Shading

- Constant Intensity or Flat Shading
- One color for the entire triangle
- Fast
- Good for some objects
- What happens if triangles are small?
- Sudden intensity changes at borders

Gouraud Shading

- Intensity Interpolation Shading
- Calculate lighting at the vertices. Then interpolate the colors as you scan convert
- Relatively fast, only do three calculations
- No sudden intensity changes



Phong Shading

- Interpolate the normal, since that is the information that represents the "curvature"
- Linearly interpolate the vertex normals. For ***each*** pixel, as you scan convert, *calculate* the lighting *per pixel*.
- True "per pixel" lighting
- Not done by most hardware/libraries/etc